

EXHALE: Convergence Reliability and the Steady-State Newton (JFNK) Solver

(work log, 2026-06-10/11)

This memo is the complete record of two tightly linked developments in EXHALE: (i) the discovery that the historical convergence criterion ($du < 10^{-3}$ on the mass-flux spread) can stop runs at premature, non-steady snapshots — the WASP-121b reference used throughout earlier work was such a snapshot, low by ~ 0.08 dex in $\dot{M}$ — together with the new convergence diagnostics that fix this; and (ii) the construction, debugging, and first converged run of a direct steady-state solver (pseudo-transient continuation + Jacobian-free Newton–Krylov), which reaches residual levels unattainable by time marching at $\sim 1/200$ of the cost. Every code change is opt-in: with all new options at their defaults, the marching code is byte-identical to the pre-existing behavior (verified by the regression harness after every increment). Statements of physical interpretation below are tentative unless explicitly verified.

Contents

1. Background and Stage-I refactor (context)	1
2. The convergence-criterion problem	2
2.1 What du measures, and what it misses	2
2.2 The premature-dip discovery (WASP-121b)	2
2.3 New diagnostics and stops (all opt-in)	2
3. The steady-state solver	3
3.1 Formulation	3
3.2 Jacobian and linear algebra	3
3.3 The five necessary ingredients (each found by a stall)	4
3.4 Result: first converged steady solve	5
4. Runtime options and hooks (reference)	5
5. Hardening results (2026-06-11, after first convergence)	6
6. Remaining open items	8
7. File inventory	9

1. Background and Stage-I refactor (context)

The work below sits on top of the Stage-I “extensibility” refactor of the same dates (named species constants; the composition single policy point for n_e/n_{tot} /EOS; schema-2 output headers and a label-mapping `load_IC` that preserves

the metal ionization state on restart; element-loop abundances; rate dispatch by ion index). Each step was a no-physics-change refactor gated by a deterministic regression harness (regression/run_check.sh: single-threaded WASP-121b runs, golden outputs compared byte-wise excluding # header lines; matrix cases wasp_full = He 2³S + metals and wasp_he23off). Details: docs/refactor_stage1_log.md. The same harness gates every solver-related change in this memo.

2. The convergence-criterion problem

2.1 What du measures, and what it misses

ATES declares steady state when

$$du = \frac{\max_j |\rho v r^2|_j - \min_j |\rho v r^2|_j}{\min_j |\rho v r^2|_j} \Big|_{j \in [j_{\min}, N]} < du_{\text{th}}, \quad du_{\text{th}} = 10^{-3}, \quad (1)$$

i.e. the *spatial spread* of the mass flux over the wind window ($r \geq r_{\text{esc}}$). Two blind spots were demonstrated this session:

1. du cannot see a *uniform level drift*: the whole $\rho v r^2$ profile can rise or fall while staying flat.
2. du is *oscillatory* on WASP-121b: it dips below 10^{-3} transiently long before the wind settles.

2.2 The premature-dip discovery (WASP-121b)

The single-thread regression golden (He 2³S + metals, two-stage PLM→WENO3, stopped by $du < 10^{-3}$ at step 7,288) has raw $\log_{10} 4\pi \rho v r^2(r_{\text{max}}) = 13.631$. Suppressing the du stop and marching on, the flux level rises *monotonically* (the windowed level change lev_rel falls $0.23 \rightarrow 2.7 \times 10^{-3}$) toward $\log_{10} \dot{M} \simeq 13.71$: at step 35,085 the state reaches 13.707 and is still creeping upward, with T_{max} falling from 11,021 K to 10,767 K. Three independent paths agree on the deeper state (long cold marching →13.707; warm restarts →13.716; the naive local- dt experiment →13.716 in 1,702 steps). **The old golden was a premature du -dip snapshot, not the steady state** — approximately 0.08 dex ($\sim 20\%$) low in $\dot{M}$.

Two earlier conclusions were corrected as a result. First, naive cell-by-cell pseudo-time stepping (“Time stepping: Local”) had been judged “answer-shifting”; in fact it reaches the *correct deeper* state $\sim 20\times$ faster — the judgment was an artifact of comparing against the premature golden. (The LTS state does carry larger hydro residuals in each cell, §??, so it is “fast but needs polishing.”) Second, the apparent “path-dependent quasi-steady states” largely collapse to one true state plus premature snapshots along the way.

2.3 New diagnostics and stops (all opt-in)

Level-stability gate (Level_tol:). A converged/stalled stop additionally requires the mean $|\rho v r^2|$ over $[j_{\min}, N]$ to change by less than lev_th (relative) across the last N_{stall} steps (ring buffer in EXHALE_main; the relative level change is now the 4th column of the stdout at each step). This correctly refuses the premature dip (the cold run marches past step 7,288), but a practical tolerance cannot discriminate ultra-slow drifts; it is necessary, not sufficient. Default off.

Steady-residual diagnostic (EXHALE_RESIDUAL=1). The finite-volume steady residual

$$\mathbf{R}_j = \begin{cases} (\nabla \cdot \mathbf{F})_j - S_j, & \text{mass, momentum,} \\ (\nabla \cdot \mathbf{F}_E)_j - S_{E,j} - (\mathcal{H}_j - \mathcal{C}_j), & \text{energy,} \end{cases} \quad (2)$$

evaluated once on a loaded state (one Reconstruct+RK_rhs pass; heating/cooling from ioniz_eq; reported as $\max_j |\mathbf{R}| / \max_j |u|$ per component over the window). This is *independent* of du/dt /stall and ranks candidate states:

state	R_{mass}	R_{mom}	R_{energy}
golden 13.631 ($du < 10^{-3}$ stop)	9.9×10^{-2}	4.7×10^{-1}	3.1×10^1
warm 13.716 (local-dt + cont.)	2.4	2.8	7.4×10^{-1}
cold 13.707 (35k steps)	2.1×10^{-2}	1.6×10^{-2}	1.5×10^{-2}

The premature golden has an energy residual of ~ 30 : heating and cooling are grossly out of balance, which du cannot detect. (The energy value 1.5×10^{-2} for the cold state corresponds to the consistent ghost-handling order, §3.1.)

Residual-based stop (Resid tol:). In stage 2 the run converges on $\max_k \max_j |\mathbf{R}_{j,k}| / \max_j |u_k| < \text{resid_th}$ (evaluated every $N_{\text{resid}}=500$ steps; cheap), *replacing* the du test. Validation: a run with $\text{resid_th} = 10^{-2}$ reached $du = 8.6 \times 10^{-4}$ (below the old stop) at step 32k and correctly kept marching because R_{max} was still 3.1×10^{-2} and falling monotonically. Default off.

3. The steady-state solver

The marching cost to a genuinely converged wind ($> 35\text{k}$ steps for WASP-121b; $> 10^6$ steps for HD189733b without reaching it) motivates solving the steady residual directly: $F(Y) = 0$.

3.1 Formulation

Unknowns $Y \in \mathbb{R}^{3N}$: the conserved hydro variables $(\rho, \rho v, E)$ on the N physical cells. Ghost cells are not unknowns; Apply_BC fills them from the interior on every residual evaluation (lower boundary: pinned ρ_{bc} , $p = 1 + dp_{\text{bc}}$ and the one-way velocity valve; upper: extrapolation). **Local elimination:** given Y , the ionization equilibrium is re-solved (ioniz_eq) and the resulting heating/cooling enter the energy residual — so F is a pure-hydro residual of size $3N$ (“method A” of the SUNDIALS plan), and the non-local radiation (column densities) is inside F . The residual construction is shared with the diagnostics (steady_residual.f90: assemble_residual); the vector form lives in steady_newton.f90 (pack_U/unpack_U/eval_residual), validated to reproduce the EXHALE_RESIDUAL numbers to all digits. One ordering subtlety: Apply_BC must run *before* the chemistry/heating step so that F depends only on the interior Y ; the diagnostic hook was aligned to the same order.

3.2 Jacobian and linear algebra

Banded approximation. With WENO3’s ± 2 -cell stencil and 3 coupled variables per cell, $\partial F_i / \partial Y_j$ has half-bandwidths $k_l = k_u = 8$ in the flat ordering. A graph coloring with stride $k_l + k_u + 1 = 17$ recovers all band entries of a *local* residual

from 17 finite-difference probes. The banded matrix is factored with LAPACK dgbtrf/dgbtrs (LDLIBS=llapack; ATLAS LAPACK 3.7.1 on this machine, link-tested). A unit test (EXHALE_JAC_TEST=1) checks Jr against a directional finite difference: relative mismatch 1.3×10^{-4} , i.e. the band layout is correct (an error would be $O(1)$).

JFNK. Because the radiation makes F weakly non-local, the banded matrix is used as a *preconditioner* while the true Jacobian-vector product is matrix-free, $Jv \simeq [F(Y + \varepsilon v) - F(Y)]/\varepsilon$ (jv_product), inside a right-preconditioned GMRES(m) (pgmres, modified Gram–Schmidt + Givens; inexact-Newton forcing 10^{-1}). The outer loop is pseudo-transient continuation,

$$\left(\frac{1}{\Delta\tau}I + J\right)\delta Y = -F(Y), \quad \Delta\tau \leftarrow \Delta\tau \max(\lambda, 0.1) \frac{\|F\|_{\text{old}}}{\|F\|_{\text{new}}} \quad (\text{SER ramp}), \quad (3)$$

with a backtracking line search and positivity guards ($\rho, p > 0$).

3.3 The five necessary ingredients (each found by a stall)

The path to convergence was a sequence of well-localized stalls; each fix was verified by its own probe (warm start from the cold-35k state, $\Delta\tau_0 = 1$ unless noted). Removing any one ingredient reproduces the corresponding stall.

1. **Smooth line-search merit.** The first driver used the component-wise max-relative residual as the merit; it is non-smooth and rejected every step ($\lambda \rightarrow 10^{-6}$). Fix: merit = $\|F\|_2$ (later $\|D^{-1}F\|_2$). After the fix the banded-PTC driver descends, but only linearly ($\sim 2\%$ /iteration) — and a $\Delta\tau_0 = 1$ probe showed a textbook Newton step (residual halved in one step) before stalling, proving the machinery and isolating the next blocker.
2. **Matrix-free Jv (JFNK).** The banded Jacobian omits the non-local radiation response; with exact Jv via GMRES the merit fell $4.5\times$ in two steps (vs. 2% /step), $\Delta\tau$ grew. Stall again — now localized (whole-domain worst-cell diagnostic) to the *base* cells $j = 1-2$ (an earlier “sonic point at r_{esc} ” reading was an artifact of searching only $[j_{\min}, N]$; the sonic point is actually at $r = 1.61$ and r_{esc} enters no physics). Freezing base cells with anchor rows does *not* help: the worst cell simply follows the slab edge and the anchor interface itself introduces a kink.
3. **Diagonal scaling.** The unknowns span ~ 7 decades (base $\rho \sim 1$ vs. wind $\sim 10^{-7}$ in code units); FD steps with absolute floors give wind cells 15–60% relative perturbations and corrupt exactly the columns where the residual lives after the first base-scale corrections. With $D_i = \max(|Y_i|, 10^{-6} \max_j |Y^{(k)}|)$, the scaled system $D^{-1}(\frac{1}{\Delta\tau}I + J)D$, D-relative FD steps, and merit $\|D^{-1}F\|_2$: GMRES needs 2–6 vectors (was 9) and the merit falls $24\times$ before stalling — at the base again.
4. **Smooth base valve.** The lower BC ghost velocity is $\max(v_1, 0)$, and the steady state hugs the kink ($v_1 \simeq +4.5 \times 10^{-3} c_s$): Newton trial steps crossing $v_1 = 0$ are non-differentiable. Opt-in replacement (Valve eps:): $v_g = \frac{1}{2}(v + \sqrt{v^2 + \varepsilon_v^2})$, smooth, $\rightarrow 0$ for $v \ll -\varepsilon_v$, $\rightarrow v$ for $v \gg \varepsilon_v$ (an earlier variant with a $-\varepsilon_v/2$ shift leaked inflow and was fixed). With $\varepsilon_v = 10^{-2}$ (too large, it perturbs the state) the solver descends $10\times$ *through* the base — differentiability is what Newton needed — but $\varepsilon_v = 10^{-4}$ still stalled at the base: the valve was not the only kink.
5. **Frozen WENO weights + non-monotone line search.** The ESWENO3 smoothness factors $S_{0,1} = 1 + \tau^2/\beta_{0,1}$ are smooth but strongly nonlinear; freezing them at the outer iterate (weno_mode: store on a mode-1 pass, reuse

on mode-2 passes for all Jacobian probes, Jv products, and line-search trials — the standard lagged-weights remedy) removes their response from the inner problem. Independently, the monotone Armijo test fails once the required decrease $10^{-4}\lambda\|F\|$ falls below the iterative-chemistry noise floor of the residual; a *non-monotone* (Grippo, memory 5) acceptance — improve on the worst of the last five accepted merits — is the standard remedy.

3.4 Result: first converged steady solve

With all five ingredients (warm start from the cold-35k state, $\Delta\tau_0 = 1$, $\varepsilon_v = 10^{-4}$):

target	iterations	final $\ R\ $
Resid tol = 10^{-3} 8 (all $\lambda = 1$, GMRES 2–3)		9.5×10^{-4}
Resid tol = 10^{-4}	22	6.0×10^{-5}

Each outer iteration costs ~ 25 residual evaluations (17 for the banded preconditioner + GMRES + line search), so the 10^{-3} solve costs the equivalent of ~ 200 marching steps — versus $> 35,000$ marching steps that never reached this residual level. Convergence is linear, not quadratic (the frozen weights are refreshed per outer iteration — a Picard–Newton hybrid — and the chemistry noise floor remains); every step is a full Newton step.

Physical validation (vs. the marching references):

state	$\log_{10} \dot{M}_{\text{raw}}$	spread($r \geq 1.5$)	T_{max}
Newton (8 its)	13.7111	5.0×10^{-4}	10,747 K
cold marching (35k steps)	13.7066 (rising)	1.2×10^{-3}	10,767 K
old golden (premature)	13.6309	1.8×10^{-3}	11,021 K

The Newton solution sits on the marching asymptote ($\simeq 13.71$), is the flattest state produced to date, and agrees with the deep-marching reference to 0.005 dex in $\dot{M}$ and 0.2% in T_{max} .

4. Runtime options and hooks (reference)

All defaults preserve the legacy behavior byte-identically (re-verified by the harness after every change in this memo).

input line	default	effect
du_th [PLM,WENO3]: a b	—	convergence thresholds; the single-stage vs. two-stage choice is set by Reconstruction scheme: (PLM/WENO3 single-stage use only a ; PLM+WENO3 two-stage: PLM until $du < a$ or stall, then WENO3 until $du < b$)
Stall [tol,N]: t n	$10^{-6}, 2000$	stall-detector override
Level tol: v	off	mass-flux level-stability gate (§2.3)
Resid tol: v	off	residual-based convergence (§2.3)
Time stepping: Local	off	cell-by-cell pseudo- dt (steady-state acceleration; reaches the deep state fast but leaves residuals in each cell)
Energy solver: Explicit	semi-implicit	original forward-Euler source update (isolation tests)
Valve eps: v	off (hard $\max(v, 0)$)	smooth base valve (§3.3)
CFL: v	0.6	CFL override

environment hook	effect
EXHALE_DUMP_IC=1	write the loaded state and stop (restart round-trip test)
EXHALE_RESIDUAL=1	evaluate and report the steady residual of the loaded state, then stop
EXHALE_NEWTON_TEST=1	pack/unpack + vector-residual self-test
EXHALE_JAC_TEST=1	banded-Jacobian vs. directional-FD unit test
EXHALE_PTC=1	run the steady solver from the (loaded) IC, write output, stop; EXHALE_PTC_JFNK=1 selects JFNK (else banded PTC); EXHALE_PTC_DTAU0, EXHALE_PTC_NFIX optional

Working recipe for the converged WASP-121b solve:

```
# input.inp: ... + "Valve eps: 1.0e-4" + "Resid tol: 1.0e-3"
# output/{Hydro_ioniz,Ion_species}_IC.txt: a warm-start state
EXHALE_PTC=1 EXHALE_PTC_JFNK=1 EXHALE_PTC_DTAU0=1.0 ./EXHALE.x
```

5. Hardening results (2026-06-11, after first convergence)

HD189733b — 33 s vs. >3 h. Warm-started from the deepest marching state available (1.16M steps, $du = 6.6 \times 10^{-3}$; starting residual $(2.4, 5.2, 2.4) \times 10^{-3}$), the JFNK solver converged to $\|R\| = 9.5 \times 10^{-4}$ in 129 iterations / **33 seconds wall time**. The converged state has $\log_{10} \dot{M}_{\text{raw}} = 10.298$ vs. 10.218 (marching) and 10.223 (pristine ATES-Code-main) — both still drifting upward — and is the flattest state produced (spread 7.6×10^{-3} vs. 1.25×10^{-2}). Tentatively, the true HD189733b $\dot{M}$ is $\sim 20\%$ above the marching estimates, the same premature-snapshot pattern quantified for WASP-121b. Tightening to 10^{-4} grinds: the frozen-weight refresh jumps are large at the breathing base and progress floors at $\|R\| \sim 3.4 \times 10^{-4}$ — the practical floor for this planet with the current settings.

Valve- ε insensitivity. Two full Newton solves of WASP-121b (tol 10^{-4}) with $\varepsilon_v = 10^{-4}$ and 10^{-5} give $\log_{10} \dot{M} = 13.7110$ and $T_{\text{max}} = 10750.2\text{ K}$ *identically* to the digits quoted: the valve smoothing is a pure numerical regularization.

Solver: Newton (production wiring) and the switch criterion. The warm-up→Newton hand-off was wired into the main loop. Two failure modes were found and fixed along the way, both reiterating the central lesson: (i) a du -based switch fails — at $du < 10^{-2}$ during the early transient the residual is still $\|R\| \approx 4.8$, far outside Newton’s basin; (ii) with the switch fixed, the marching’s own $du < 10^{-3}$ stop fired first, returning the premature state. The final design: *the switch and the warm-up termination are both residual-based* (periodic monitor; switch at $\max \|R\| < 5 \times 10^{-2}$, du -stops suspended in this mode). End-to-end validation from a *cold isothermal IC* (WASP-121b, single input line Solver: Newton): two-stage warm-up to step 27,000 ($\|R\| = 4.85 \times 10^{-2}$) → JFNK 16 iterations → $\|R\| = 5.4 \times 10^{-4}$ → standard outputs and post-processing; final $\log_{10} \dot{M}_{\text{raw}} = 13.7111$, spread 5.1×10^{-4} , $T_{\text{max}} = 10,748$ K — identical to the warm-started reference. Wall time 16 min (4 threads), dominated by the warm-up.

Transmission spectra of the Newton solutions. EXHALE_transit.py was run on both converged solutions (after PP-only runs generated the *_adv profiles): He 10830 / Ly α / H α / H β spectra for both planets plus the metal lines for WASP-121b (Mg II 2796: 3.7%, Ca II K: 3.8%, Na I D2: 1.7% line-center). Peak line-center absorptions: WASP-121b He 3.7%, H α 3.7%; HD189733b He 11.9%, Ly α 46%, H α 1.8% (consistent in magnitude with the Jensen et al. 2012 H α detection). Figures and numbers are collected in regression/steady_solver_results.ipynb; usage is documented in the updated user manual (§ TPM).

Failure containment (a genuine JFNK failure, and two fixes). Re-running the manual’s four figure runs through the production pipeline exposed a real failure mode on the spherical, metals-off tutorial planet (a generic hot Jupiter, domain to $10R_p$). The solve nearly converged ($\|R\| = 6.0 \times 10^{-3}$ at iteration 2), but the non-monotone reference still held the huge *initial* merit ($\|F_s\|_2 \simeq 10^4$), so a blow-up step was “accepted”; the search then stalled at $\lambda \simeq 10^{-6}$, accepting zero-length steps against the now-flat reference for ~ 470 iterations, and — worst of all — the main loop accepted the final unconverged state ($\|R\| = 0.42$) as converged and wrote it out. Two fixes (both in the default-off Newton path; legacy byte-identity re-verified): (i) *best-iterate tracking and fail-fast* in solve_steady_jfnk — the best residual state is stored, and the solve aborts with info=2 after 15 outer iterations without a new best, returning the best iterate; (ii) in EXHALE_main, info $\neq 0$ is treated as a failure: the run prints JFNK failed, keeps the best iterate, disables the Newton mode, and *resumes plain time-marching* with the du -based stops re-armed, so an unconverged Newton state can never masquerade as the final answer. Re-run with the fixes, the same configuration aborts after 15 stuck iterations, returns the best iterate ($\|R\| = 5.97 \times 10^{-3}$), and the resumed marching stops one step later ($du = 3.4 \times 10^{-4}$, already below threshold); the run ends on an honest, near-converged state instead of a diverged one. The three other figure runs converged cleanly through the pipeline (tutorial metals-on $\|R\| = 5.6 \times 10^{-4}$, WASP-121b spherical Case A 4.1×10^{-4} , Roche Case B 5.8×10^{-4} , all info=0).

Can JFNK start cold? (basin-boundary experiments, 2026-06-11). Three probes on WASP-121b (all with the fail-fast guard, so each failure is cheap and the run still finishes honestly via the marching fallback):

start of JFNK	start $\ R\ $	JFNK outcome	final answer
cold isothermal IC	1.4×10^4	1 iter $\rightarrow$ 11, hard stall at the base	— (abort)
step 6,500 ($R_{\text{switch}}=0.5$)	0.43	$\rightarrow$ 0.245, stall	fallback <i>du</i> -stop, $\dot{M}$ 13.34 (premature)
step 12,500 ($R_{\text{switch}}=0.1$)	0.091	first step <i>worsens</i> (0.148), stall	fallback <i>du</i> -stop, $\dot{M}$ 13.38 (premature)
step 27,000 (default 5×10^{-2})	0.0485	16 iters $\rightarrow 5.4 \times 10^{-4}$	$\dot{M}$ 13.7111 ✓

The Newton basin boundary on this planet therefore lies between $\|R\| \simeq 0.09$ and 0.049 — the default switch is calibrated essentially at the edge, and raising it buys nothing: every stall pins to the first base cells ($j = 1\text{--}2$, momentum), where the valve/ionization coupling makes the frozen-weight local model invalid far from the fixed point. PTC’s small- $\Delta\tau$ limit is just implicit time stepping at $\sim 20\text{--}60\times$ the cost per unit pseudo-time of an explicit RK step, so a cold-start PTC could not beat the marching warm-up economically even where it survives. Practical caveat: when a Solver: Newton run prints JFNK failed, the fallback finishes on the *du* stop, which can be a premature state (13.34/13.38 vs. 13.71 here) — re-check such runs with EXHALE_RESIDUAL=1. Probes preserved in regression/{jfnk_cold, newton_rsw05, newton_rsw01}.

6. Remaining open items

1. **HD189733b residual floor.** Progress floors at $\|R\| \sim 3.4 \times 10^{-4}$ (frozen-weight refresh jumps at the breathing base); an under-relaxed weight refresh between outer iterations is the natural next step.
2. **Non-monotone reference poisoning.** The tutorial-planet failure above was *contained*, not prevented: the 5-entry Grippo memory can retain a huge early merit and thereby accept a blow-up step. Capping the reference at a multiple of the current merit (or flushing the history after a large drop) should prevent it outright.
3. **Chemistry noise floor.** The non-monotone search works around the `ioniz_eq` iteration noise; tightening the inner solve (or smoothing its switch points) would restore certifiable descent and possibly quadratic tails.
4. **Quasi-steady physics.** The deep states still show slow residual decay rather than an exact fixed point (base breathing); whether an exact steady state of the discrete scheme exists with the hard valve remains open — the smooth-valve fixed point found here is the natural regularized answer.

7. File inventory

file	content
src/modules/time_step/steady_residual.f90	shared steady-residual assembly + norms (diagnostics, monitor, Newton)
src/modules/time_step/steady_newton.f90	pack/unpack, residuals, scaling, banded Jacobians (frozen/full), jv_product, pgmres, solve_steady_ptc, solve_steady_jfnk, base-anchor option
src/modules/states/Reconstruction.f90	weno_mode frozen-weights support (S0/S1 store/reuse)
src/modules/states/Apply_BC.f90	smooth base valve (valve_eps)
src/modules/init/parameters.f90	new option variables (defaults off)
src/modules/files_IO/input_read.f90	option parsing (keyword lines after Force start:)
EXHALE_main.f90	level/residual monitors and stops; env hooks
regression/	harness, goldens, and all probe run directories (ptc_warm, crit_cold, resid_*, ...)
docs/steady_solver_design.md	incremental design notes (§§4–11) with the raw probe logs
docs/refactor_stage1_log.md	Stage-I extensibility change log